Ashutosh Kumar Jha

Bengaluru, Karnataka, India
LinkedIn: ashutosh-kumar-jha
GitHub: github.com/ashutosh2080
Email: jhaashutosh470@gmail.com
Mobile: +91-9482967870
Seeking: Hardware / Embedded Internships

Profile Summary

B.Tech Computer Science and Engineering student with hands-on project experience in Arduino- and ESP32-based devices, hardware prototyping, and 3D printing. Interested in embedded systems, IoT, robotics, and electronics, with a research-oriented approach to exploring topics from multiple perspectives and comparing possible solutions.

Technical Skills

Programming	C, Python, JavaScript
Embedded Platforms	Arduino, ESP32, ESP8266
Hardware / Prototyping	Sensors, microcontrollers, multimeter use, soldering
CAD / 3D Design	Onshape, Tinkercad
3D Printing Software	Ultimaker Cura, Anycubic Slicer, Pronterface
Development Environment	Arduino IDE

Education

Jain University — 2025 – Present
B.Tech in Computer Science and Engineering — Expected Graduation: 2029
CGPA: 8.5 — Bengaluru, Karnataka

Gyanda Academy — Birgunj, Nepal
Board: NEB — Percentage: 79.35%

South Zone Secondary School — Birgunj, Nepal
Board: NEB — Percentage: 85%

Experience

Robotics Educator and Teacher Trainer — 2023–2025

- Worked as a contract-based robotics teacher in schools, teaching robotics concepts through practical, hands-on learning activities.
- Trained approximately 4–5 teachers and taught 400–600 students across schools, colleges, and educational institutions through lessons, workshops, and demonstrations.

Projects

- **3D Printer Built from Scratch**
  Designed and assembled a 3D printer from scratch, combining mechanical assembly with a 3D-printing workflow using CAD and printer software.
- **Arduino-Based Emergency Safety Device (GPS + GSM)**
  Designed an emergency safety device intended to obtain GPS location data and send location information during an emergency using GSM communication.
- **Internet-Connected ESP32 Table Bot with Web API**
  Built an internet-connected ESP32 table bot with an API interface for two-way live data exchange between the device and a web application.

Other Hardware Projects

- **ESP32 Wi-Fi-Controlled Car:** Built a car controlled over Wi-Fi using ESP32.
- **ESP32 Home Automation:** Developed a home-automation project using ESP32.
- **Automatic Institutional Bell:** Created an Arduino-based automatic bell system for institutional use.
- **Agriculture Bot:** Built an Arduino-based bot intended for agricultural applications.

Achievements

- **Organizer, ORBIT 2.0 International Expo:** One of 30 organizers for Nepal's first international engineering expo while in Class XII.
- **3D Printer Recognition:** Built a 3D printer from scratch in Class X; recognized by my school and featured by Routine of Nepal Banda, local news networks, and a local newspaper.
- **Robotics Outreach:** Worked with multiple institutions in my hometown to promote robotics awareness and conduct workshops and exhibitions.

Additional Information

Target Roles: Hardware, embedded systems, IoT, robotics, and electronics internships
Availability: 4–6 hours per day
Work Preference: Open to remote and other suitable arrangements